

ADESOLA KAREEM

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PROFESSIONAL SUMMARY

Backend Software Engineer with 2+ years of production Python experience, specializing in distributed pipelines, high-throughput data systems, and LLM-powered platforms. At WebMD, owned the full Athena vector ingestion pipeline — designed pgvector storage, resolved 6 performance bottlenecks under a 2M-document load test, and drove deployments across DevInt, QA, Staging, Production, and HIPAA environments. Strong foundation in system design, API architecture, and observability.

SKILLS

Languages: Python (5+ yrs), JavaScript/Node.js, Java

Backend: FastAPI, REST APIs, Pydantic, JSON Schema, Express.js, Microservices, Event-Driven Architecture

Data: PostgreSQL (pgvector), Redis, MongoDB, Solr

Infra: RabbitMQ, Docker, GCP, GitLab CI/CD, JWT/OAuth2

AI/LLM: LLM routing, embeddings (OpenAI, Azure, Anthropic, Vertex, Cohere), RAG, FAISS, PII anonymization

WORK EXPERIENCE

WebMD — Software Engineer (Backend)

Mar 2025 – Present | Python · FastAPI · RabbitMQ · pgvector · Redis · Docker

- Resolved 6 concurrent performance bottlenecks under a 2M-document load test: async double-checked locking, TCP connection reuse (eliminating per-request TLS handshakes), background task GC safety, bulk DELETE CTE, connection reference cleanup, and logger crash fix
- Reduced DB round-trips to 3 for any batch size via bulk unnest() inserts (bulk delete + doc insert + chunk+embedding insert)
- Integrated tenacity retry logic (3 attempts, exponential backoff 1s–8s) for transient 502/503/504 errors from embeddings API — eliminated pod restarts under load
- Built pgvector storage backend from scratch: asyncpg, HNSW index, schema-per-business-unit multi-tenant routing, and GIN full-text index on chunk content
- Implemented batch processing pipeline: in-memory buffer with configurable flush on BATCH_SIZE or interval; dedup key across 6 composite fields for idempotent processing
- Designed multi-backend storage routing (pgvector + Solr) by business unit via env config, with per-backend failure isolation

Nomba — Software Engineer

Nov 2023 – Feb 2024 | Python · MongoDB · Redis · GCP

- Reduced GCP infrastructure costs by 25% via caching and service optimization
- Built a CSV export pipeline using MongoDB and Redis

Precious Cornerstone University (NYSC) — Research Assistant

Mar 2024 – Jan 2025

- Trained faculty on AI tools (ChatGPT, Claude), improving team productivity by ~40%

PROJECTS

Scalable Notification Service — Apr 2026 – Present

FastAPI · RabbitMQ · Redis · PostgreSQL · Docker · Prometheus · Grafana | github.com/kareemadesola/scalable-notification-service

- Production-grade multi-channel notification system (email, SMS, push, WebSocket) with independent channel scaling
- Implemented exponential backoff retry + dead-letter queues for fault tolerance
- Built Redis rate limiting and Prometheus + Grafana observability (latency, throughput, error tracking)

AI Investigation & Research Assistant — Apr 2026

Python · FAISS · RAG · Agentic pipeline | github.com/kareemadesola/ai-investigation-assistant

- Agentic RAG system combining vector search, multi-step reasoning, and structured evaluation
- Built hallucination detection and response consistency evaluation layer

EDUCATION

Obafemi Awolowo University, Ile-Ife, Nigeria — Aug 2023

BSc Computer Science with Mathematics — First Class Honours

CERTIFICATIONS

HackerRank: Java (2024), REST API (2023), Problem Solving (2023), Python (2022)